HCL TECH - Hackathon
Use case 3 : Healthcare Analytics & Patients Insights

Problem statement :

use patient data to improve care, predict risks, and optimize operation

Process :

1. Database + Schema
2. Setup Stage + File format
3. Raw tables
4. COPY INTO (load data)
5. Streams (incremental tracking)
6. Validation layer (data quality)
7. Curated layer (transform + SCD2)
8. Data mart (business tables)
9. KPI queries
10. Tasks (automation)

Program :

```
CREATE OR REPLACE DATABASE HEALTHCARE_ADVANCED_DB;
USE DATABASE HEALTHCARE_ADVANCED_DB;

CREATE OR REPLACE WAREHOUSE HEALTHCARE_WH
WITH WAREHOUSE_SIZE='X-SMALL'
AUTO_SUSPEND=60
AUTO_RESUME=TRUE ;
USE WAREHOUSE HEALTHCARE_WH;

CREATE OR REPLACE SCHEMA raw;
CREATE OR REPLACE SCHEMA validated;
CREATE OR REPLACE SCHEMA curated;
CREATE OR REPLACE SCHEMA external_stages;
CREATE OR REPLACE SCHEMA governance;
CREATE OR REPLACE SCHEMA datamart;
CREATE OR REPLACE SCHEMA analytical;

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-- 1. RAW LAYER & INGESTION
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CREATE OR REPLACE FILE FORMAT raw.csv_format
  TYPE = 'CSV'
  FIELD_DELIMITER = ','
  SKIP_HEADER = 1
  NULL_IF = ('', 'NULL', 'null')
  EMPTY_FIELD_AS_NULL = TRUE
  FIELD_OPTIONALLY_ENCLOSED_BY = '"';

CREATE OR REPLACE STAGE external_stages.healthcare_stage;

-- Raw Tables
CREATE OR REPLACE TABLE raw.raw_patients (
    patient_id VARCHAR,
    first_name VARCHAR,
    last_name VARCHAR,
    dob DATE,
    gender VARCHAR,
    blood_group VARCHAR,
    city VARCHAR,
    state VARCHAR,
    registration_date TIMESTAMP_NTZ,
    _load_timestamp TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
);

CREATE OR REPLACE TABLE raw.raw_appointments (
    appointment_id VARCHAR,
    patient_id VARCHAR,
    appointment_date TIMESTAMP_NTZ,
    department VARCHAR,
    doctor_id VARCHAR,
    status VARCHAR,
    _load_timestamp TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
);

CREATE OR REPLACE TABLE raw.raw_medical_records (
    record_id VARCHAR,
    patient_id VARCHAR,
    diagnosis VARCHAR,
    heart_rate VARCHAR, -- Read as varchar to catch bad cast
    blood_pressure_systolic VARCHAR,
    blood_pressure_diastolic VARCHAR,
    cholesterol_level VARCHAR,
    record_date TIMESTAMP_NTZ,
    _load_timestamp TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
);

CREATE OR REPLACE TABLE raw.raw_billing (
    bill_id VARCHAR,
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```
    patient_id VARCHAR,
    appointment_id VARCHAR,
    total_amount VARCHAR,
    insurance_covered VARCHAR,
    patient_paid VARCHAR,
    billing_date TIMESTAMP_NTZ,
    status VARCHAR,
    _load_timestamp TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
);


-- Streams
CREATE OR REPLACE STREAM raw.patients_stream ON TABLE raw.raw_patients;
CREATE OR REPLACE STREAM raw.appointments_stream ON TABLE raw.raw_appointments;
CREATE OR REPLACE STREAM raw.medical_records_stream ON TABLE raw.raw_medical_records;
CREATE OR REPLACE STREAM raw.billing_stream ON TABLE raw.raw_billing;

-- COPY INTO Commands (Execute these after uploading CSVs to @external_stages.healthcare_stage)
COPY INTO raw.raw_patients (patient_id, first_name, last_name, dob, gender, blood_group, city, state, registration_date)
FROM @external_stages.healthcare_stage
FILE_FORMAT = raw.csv_format
PATTERN = '.*patients.*\.csv'
ON_ERROR = CONTINUE;

COPY INTO raw.raw_appointments (appointment_id, patient_id, appointment_date, department, doctor_id, status)
FROM @external_stages.healthcare_stage
FILE_FORMAT = raw.csv_format
PATTERN = '.*appointments.*\.csv'
ON_ERROR = CONTINUE;

COPY INTO raw.raw_medical_records (record_id, patient_id, diagnosis, heart_rate, blood_pressure_systolic, blood_pressure_diastolic, cholesterol_level, record_date)
FROM @external_stages.healthcare_stage
FILE_FORMAT = raw.csv_format
PATTERN = '.*medical_records.*\.csv'
ON_ERROR = CONTINUE;

COPY INTO raw.raw_billing (bill_id, patient_id, appointment_id, total_amount, insurance_covered, patient_paid, billing_date, status)
FROM @external_stages.healthcare_stage
FILE_FORMAT = raw.csv_format
PATTERN = '.*billing.*\.csv'
ON_ERROR = CONTINUE;
```

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-- 2. VALIDATED LAYER (Data Cleaning & DQ Logging)
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CREATE OR REPLACE TABLE validated.valid_patients (
    patient_id VARCHAR PRIMARY KEY,
    first_name VARCHAR,
    last_name VARCHAR,
    dob DATE,
    gender VARCHAR,
    blood_group VARCHAR,
    city VARCHAR,
    state VARCHAR,
    registration_date TIMESTAMP_NTZ,
    _load_timestamp TIMESTAMP_NTZ
);

CREATE OR REPLACE TABLE validated.valid_appointments (
    appointment_id VARCHAR PRIMARY KEY,
    patient_id VARCHAR,
    appointment_date TIMESTAMP_NTZ,
    department VARCHAR,
    doctor_id VARCHAR,
    status VARCHAR,
    _load_timestamp TIMESTAMP_NTZ
);

CREATE OR REPLACE TABLE validated.valid_medical_records (
    record_id VARCHAR PRIMARY KEY,
    patient_id VARCHAR,
    diagnosis VARCHAR,
    heart_rate NUMBER,
    blood_pressure_systolic NUMBER,
    blood_pressure_diastolic NUMBER,
    cholesterol_level NUMBER,
    record_date TIMESTAMP_NTZ,
    _load_timestamp TIMESTAMP_NTZ
);

CREATE OR REPLACE TABLE validated.valid_billing (
    bill_id VARCHAR PRIMARY KEY,
    patient_id VARCHAR,
    appointment_id VARCHAR,
    total_amount FLOAT,
```

```
    insurance_covered FLOAT,
    patient_paid FLOAT,
    billing_date TIMESTAMP_NTZ,
    status VARCHAR,
    _load_timestamp TIMESTAMP_NTZ
);

-- Governance Exception Tables
CREATE OR REPLACE TABLE governance.dq_exception_log (
    error_id VARCHAR DEFAULT UUID_STRING(),
    source_table VARCHAR,
    business_key VARCHAR,
    error_type VARCHAR,
    error_message VARCHAR,
    _recorded_at TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
);

CREATE OR REPLACE TABLE governance.anomaly_rule_hits (
    anomaly_id VARCHAR DEFAULT UUID_STRING(),
    rule_name VARCHAR,
    business_key VARCHAR,
    anomaly_desc VARCHAR,
    detected_at TIMESTAMP_NTZ DEFAULT CURRENT_TIMESTAMP()
);

-- Stored Procedure for DQ and Merging
CREATE OR REPLACE PROCEDURE validated.process_data_quality()
RETURNS STRING
LANGUAGE SQL
AS
$$
BEGIN
    BEGIN TRANSACTION;
    -- 1. PROCESS PATIENTS
            CREATE TEMPORARY TABLE cur_patients_stream AS SELECT * FROM raw.patients_stream WHERE METADATA$ACTION = 'INSERT';

          INSERT INTO governance.dq_exception_log (source_table, business_key, error_type, error_message)
    SELECT 'raw_patients', patient_id, 'NULL_VALUES', 'Missing Gender or City'
    FROM cur_patients_stream WHERE gender IS NULL OR city IS NULL;

    MERGE INTO validated.valid_patients tgt
    USING (
       SELECT * FROM (
                    SELECT *, ROW_NUMBER() OVER(PARTITION BY patient_id ORDER BY _load_timestamp DESC) as rn
          FROM cur_patients_stream WHERE gender IS NOT NULL AND city IS NOT NULL
```

```
    ) WHERE rn = 1
  ) src ON tgt.patient_id = src.patient_id
        WHEN MATCHED THEN UPDATE SET tgt.first_name = UPPER(src.first_name), tgt.last_name = UPPER(src.last_name), tgt.city = UPPER(src.city), tgt.state = UPPER(src.state), tgt.gender = src.gender, tgt.blood_group = src.blood_group
     WHEN NOT MATCHED THEN INSERT (patient_id, first_name, last_name, dob, gender, blood_group, city, state, registration_date, _load_timestamp)
    VALUES (src.patient_id, UPPER(src.first_name), UPPER(src.last_name), src.dob, src.gender, src.blood_group, UPPER(src.city), UPPER(src.state), src.registration_date, src._load_timestamp);

  -- 2. PROCESS APPOINTMENTS
        CREATE TEMPORARY TABLE cur_appointments_stream AS SELECT * FROM raw.appointments_stream WHERE METADATA$ACTION = 'INSERT';

        INSERT INTO governance.dq_exception_log (source_table, business_key, error_type, error_message)
   SELECT 'raw_appointments', appointment_id, 'INVALID_STATUS', CONCAT('Status ', status, ' is invalid')
  FROM cur_appointments_stream WHERE status NOT IN ('Scheduled', 'Completed', 'No-Show');

  MERGE INTO validated.valid_appointments tgt
  USING (
    SELECT * FROM (
          SELECT *, ROW_NUMBER() OVER(PARTITION BY appointment_id ORDER BY _load_timestamp DESC) as rn
      FROM cur_appointments_stream
      WHERE status IN ('Scheduled', 'Completed', 'No-Show')
       AND patient_id IN (SELECT patient_id FROM validated.valid_patients)
    ) WHERE rn = 1
  ) src ON tgt.appointment_id = src.appointment_id
     WHEN MATCHED THEN UPDATE SET tgt.status = src.status, tgt.appointment_date = src.appointment_date
     WHEN NOT MATCHED THEN INSERT (appointment_id, patient_id, appointment_date, department, doctor_id, status, _load_timestamp)
    VALUES (src.appointment_id, src.patient_id, src.appointment_date, UPPER(src.department), src.doctor_id, src.status, src._load_timestamp);

  -- 3. PROCESS MEDICAL RECORDS
        CREATE TEMPORARY TABLE cur_medical_stream AS SELECT * FROM raw.medical_records_stream WHERE METADATA$ACTION = 'INSERT';

        INSERT INTO governance.dq_exception_log (source_table, business_key, error_type, error_message)
     SELECT 'raw_medical_records', record_id, 'DATA_TYPE_ERROR', 'Non-numeric clinical values'
  FROM cur_medical_stream
   WHERE TRY_CAST(cholesterol_level AS NUMBER) IS NULL OR TRY_CAST(heart_rate AS NUMBER) IS NULL;
```

```
MERGE INTO validated.valid_medical_records tgt
USING (
  SELECT * FROM (
            SELECT *, ROW_NUMBER() OVER(PARTITION BY record_id ORDER BY _load_timestamp DESC) as rn
    FROM cur_medical_stream
    WHERE TRY_CAST(cholesterol_level AS NUMBER) IS NOT NULL
     AND TRY_CAST(heart_rate AS NUMBER) IS NOT NULL
     AND patient_id IN (SELECT patient_id FROM validated.valid_patients)
  ) WHERE rn = 1
) src ON tgt.record_id = src.record_id
   WHEN MATCHED THEN UPDATE SET tgt.diagnosis = src.diagnosis, tgt.heart_rate = TRY_CAST(src.heart_rate AS NUMBER)
    WHEN NOT MATCHED THEN INSERT (record_id, patient_id, diagnosis, heart_rate, blood_pressure_systolic, blood_pressure_diastolic, cholesterol_level, record_date, _load_timestamp)
VALUES (src.record_id, src.patient_id, src.diagnosis, TRY_CAST(src.heart_rate AS NUMBER), TRY_CAST(src.blood_pressure_systolic AS NUMBER), TRY_CAST(src.blood_pressure_diastolic AS NUMBER), TRY_CAST(src.cholesterol_level AS NUMBER), src.record_date, src._load_timestamp);

-- 4. PROCESS BILLING
 CREATE TEMPORARY TABLE cur_billing_stream AS SELECT * FROM raw.billing_stream WHERE METADATA$ACTION = 'INSERT';

    INSERT INTO governance.dq_exception_log (source_table, business_key, error_type, error_message)
SELECT 'raw_billing', bill_id, 'VALUE_ERROR', 'Negative or missing amount'
    FROM cur_billing_stream WHERE TRY_CAST(total_amount AS FLOAT) < 0 OR TRY_CAST(total_amount AS FLOAT) IS NULL;

MERGE INTO validated.valid_billing tgt
USING (
  SELECT * FROM (
              SELECT *, ROW_NUMBER() OVER(PARTITION BY bill_id ORDER BY _load_timestamp DESC) as rn
    FROM cur_billing_stream
    WHERE TRY_CAST(total_amount AS FLOAT) >= 0
     AND appointment_id IN (SELECT appointment_id FROM validated.valid_appointments)
  ) WHERE rn = 1
) src ON tgt.bill_id = src.bill_id
    WHEN MATCHED THEN UPDATE SET tgt.status = src.status, tgt.total_amount = TRY_CAST(src.total_amount AS FLOAT)
  WHEN NOT MATCHED THEN INSERT (bill_id, patient_id, appointment_id, total_amount, insurance_covered, patient_paid, billing_date, status, _load_timestamp)
```

```
        VALUES (src.bill_id, src.patient_id, src.appointment_id, TRY_CAST(src.total_amount AS FLOAT), TRY_CAST(src.insurance_covered AS FLOAT), TRY_CAST(src.patient_paid AS FLOAT), src.billing_date, src.status, src._load_timestamp);

   COMMIT;
   RETURN 'Validation and Insert into VALIDATED layer completed successfully.';
EXCEPTION
   WHEN OTHER THEN
      ROLLBACK;
      RETURN 'Error in process_data_quality: ' || SQLERRM;
END;
$$;


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-- 3. CURATED LAYER (Dimensions & Facts with Surrogate Keys)
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-- SCD-2 Patient Dimension
CREATE OR REPLACE TABLE curated.dim_patient (
   patient_key STRING,
   patient_id STRING,
   full_name STRING,
   dob DATE,
   gender STRING,
   blood_group STRING,
   city STRING,
   state STRING,
   registration_date TIMESTAMP_NTZ,
   effective_date DATE,
   end_date DATE,
   is_current BOOLEAN
);

-- Fact Appointments (Event Fact)
CREATE OR REPLACE TABLE curated.fact_appointments (
   appointment_key STRING,
   appointment_id STRING,
   patient_key STRING,
   doctor_id STRING,
   department STRING,
   appointment_date TIMESTAMP_NTZ,
   status STRING
);
```

```
-- Fact Medical Records (Metrics Fact)
CREATE OR REPLACE TABLE curated.fact_medical_records (
    record_key STRING,
    record_id STRING,
    patient_key STRING,
    diagnosis STRING,
    heart_rate NUMBER,
    blood_pressure_systolic NUMBER,
    blood_pressure_diastolic NUMBER,
    cholesterol_level NUMBER,
    record_date TIMESTAMP_NTZ
);

-- Fact Billing (Financial Fact)
CREATE OR REPLACE TABLE curated.fact_billing (
    billing_key STRING,
    bill_id STRING,
    appointment_key STRING,
    patient_key STRING,
    total_amount FLOAT,
    insurance_covered FLOAT,
    patient_paid FLOAT,
    billing_date TIMESTAMP_NTZ,
    status STRING
);


CREATE OR REPLACE PROCEDURE curated.populate_curated_models()
RETURNS STRING
LANGUAGE SQL
AS
$$
BEGIN
    BEGIN TRANSACTION;
    -- 1. SCD-2 for PATIENT DIMENSION
    UPDATE curated.dim_patient tgt
    SET tgt.end_date = CURRENT_DATE(), tgt.is_current = FALSE
    FROM validated.valid_patients src
    WHERE tgt.patient_id = src.patient_id
      AND tgt.is_current = TRUE
      AND (tgt.city != src.city OR tgt.state != src.state);

    MERGE INTO curated.dim_patient tgt
    USING validated.valid_patients src
    ON tgt.patient_id = src.patient_id AND tgt.is_current = TRUE
    WHEN NOT MATCHED THEN
```

```
INSERT (patient_key, patient_id, full_name, dob, gender, blood_group, city, state, registration_date, effective_date, end_date, is_current)
VALUES (UUID_STRING(), src.patient_id, CONCAT(src.first_name, ' ', src.last_name), src.dob, src.gender, src.blood_group, src.city, src.state, src.registration_date, CURRENT_DATE(), NULL, TRUE);

-- 2. FACT APPOINTMENTS
MERGE INTO curated.fact_appointments tgt
USING (
    SELECT
      a.appointment_id,
      p.patient_key,
      a.doctor_id,
      a.department,
      a.appointment_date,
      a.status
    FROM validated.valid_appointments a
    JOIN curated.dim_patient p ON a.patient_id = p.patient_id AND p.is_current = TRUE
) src ON tgt.appointment_id = src.appointment_id
WHEN MATCHED THEN UPDATE SET tgt.status = src.status
    WHEN NOT MATCHED THEN INSERT (appointment_key, appointment_id, patient_key, doctor_id, department, appointment_date, status)
    VALUES (UUID_STRING(), src.appointment_id, src.patient_key, src.doctor_id, src.department, src.appointment_date, src.status);

-- 3. FACT MEDICAL RECORDS
MERGE INTO curated.fact_medical_records tgt
USING (
    SELECT
      m.record_id,
      p.patient_key,
      m.diagnosis,
      m.heart_rate,
      m.blood_pressure_systolic,
      m.blood_pressure_diastolic,
      m.cholesterol_level,
      m.record_date
    FROM validated.valid_medical_records m
    JOIN curated.dim_patient p ON m.patient_id = p.patient_id AND p.is_current = TRUE
) src ON tgt.record_id = src.record_id
    WHEN MATCHED THEN UPDATE SET tgt.diagnosis = src.diagnosis, tgt.heart_rate = src.heart_rate
    WHEN NOT MATCHED THEN INSERT (record_key, record_id, patient_key, diagnosis, heart_rate, blood_pressure_systolic, blood_pressure_diastolic, cholesterol_level, record_date)
    VALUES (UUID_STRING(), src.record_id, src.patient_key, src.diagnosis, src.heart_rate, src.blood_pressure_systolic, src.blood_pressure_diastolic, src.cholesterol_level, src.record_date);

-- 4. FACT BILLING
```

```
    MERGE INTO curated.fact_billing tgt
    USING (
        SELECT
          b.bill_id,
          fa.appointment_key,
          p.patient_key,
          b.total_amount,
          b.insurance_covered,
          b.patient_paid,
          b.billing_date,
          b.status
        FROM validated.valid_billing b
        JOIN curated.dim_patient p ON b.patient_id = p.patient_id AND p.is_current = TRUE
        LEFT JOIN curated.fact_appointments fa ON b.appointment_id = fa.appointment_id
    ) src ON tgt.bill_id = src.bill_id
            WHEN MATCHED THEN UPDATE SET tgt.status = src.status, tgt.patient_paid =
src.patient_paid
      WHEN NOT MATCHED THEN INSERT (billing_key, bill_id, appointment_key, patient_key,
total_amount, insurance_covered, patient_paid, billing_date, status)
     VALUES (UUID_STRING(), src.bill_id, src.appointment_key, src.patient_key, src.total_amount,
src.insurance_covered, src.patient_paid, src.billing_date, src.status);

    COMMIT;
    RETURN 'Curated models populated successfully.';
EXCEPTION
    WHEN OTHER THEN
        ROLLBACK;
        RETURN 'Error in populate_curated_models: ' || SQLERRM;
END;
$$;


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-- 4. GOVERNANCE (Anomaly Engine)
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CREATE OR REPLACE PROCEDURE governance.run_anomaly_engine()
RETURNS STRING
LANGUAGE SQL
AS
$$
BEGIN
    BEGIN TRANSACTION;
    -- Duplicate Appointments (Patient booked twice in same day)
    INSERT INTO governance.anomaly_rule_hits (rule_name, business_key, anomaly_desc)
```

```
            SELECT 'DOUBLE_BOOKED', patient_key, CONCAT('Patient booked twice on ',
DATE(appointment_date))
   FROM curated.fact_appointments
   WHERE status != 'No-Show'
   GROUP BY patient_key, DATE(appointment_date) HAVING COUNT(appointment_id) > 1;

   -- Extreme Billing Outliers (> 3 Standard Deviations)
   INSERT INTO governance.anomaly_rule_hits (rule_name, business_key, anomaly_desc)
   WITH DeptAvg AS (
            SELECT a.department, AVG(b.total_amount) as avg_amt, STDDEV(b.total_amount) as
std_dev
          FROM curated.fact_billing b JOIN curated.fact_appointments a ON b.appointment_key =
a.appointment_key GROUP BY a.department
   )
    SELECT 'BILL_OUTLIER', b.bill_id, CONCAT('Amount ', b.total_amount, ' exceeds dept avg ',
d.avg_amt)
   FROM curated.fact_billing b
   JOIN curated.fact_appointments a ON b.appointment_key = a.appointment_key
   JOIN DeptAvg d ON a.department = d.department
   WHERE b.total_amount > (d.avg_amt + (3 * COALESCE(d.std_dev, 1)));

   COMMIT;
   RETURN 'Anomaly Engine run executed.';
EXCEPTION
   WHEN OTHER THEN
      ROLLBACK;
      RETURN 'Error in run_anomaly_engine: ' || SQLERRM;
END;
$$;


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==================================================================================
==
-- 5. ORCHESTRATION (Tasks)
-                                                                                                    -
==================================================================================
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CREATE OR REPLACE TASK analytical.task_process_dq
   WAREHOUSE = HEALTHCARE_WH
   SCHEDULE = '1 MINUTE'
   WHEN SYSTEM$STREAM_HAS_DATA('RAW.PATIENTS_STREAM')
     OR SYSTEM$STREAM_HAS_DATA('RAW.APPOINTMENTS_STREAM')
     OR SYSTEM$STREAM_HAS_DATA('RAW.MEDICAL_RECORDS_STREAM')
     OR SYSTEM$STREAM_HAS_DATA('RAW.BILLING_STREAM')
AS
```

```
    CALL validated.process_data_quality();

CREATE OR REPLACE TASK analytical.task_populate_curated
    WAREHOUSE = HEALTHCARE_WH
    AFTER analytical.task_process_dq
AS
    CALL curated.populate_curated_models();

CREATE OR REPLACE TASK analytical.task_anomaly_engine
    WAREHOUSE = HEALTHCARE_WH
    AFTER analytical.task_populate_curated
AS
    CALL governance.run_anomaly_engine();

ALTER TASK analytical.task_anomaly_engine RESUME;
ALTER TASK analytical.task_populate_curated RESUME;
ALTER TASK analytical.task_process_dq RESUME;


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-- 6. DATAMART LAYER (Solving the Prompt Requests)
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-- Requirement 1: Patient 360
CREATE OR REPLACE VIEW datamart.patient_360_mart AS
SELECT
    p.patient_key,
    p.patient_id,
    p.full_name,
    DATEDIFF('year', p.dob, CURRENT_DATE()) AS age,
    p.gender,
    p.city,
    COUNT(DISTINCT a.appointment_id) AS lifetime_appointments,
    COUNT(DISTINCT m.record_id) AS lifetime_medical_records,
    SUM(b.total_amount) AS lifetime_billing,
    SUM(b.patient_paid) AS lifetime_out_of_pocket
FROM curated.dim_patient p
LEFT JOIN curated.fact_appointments a ON p.patient_key = a.patient_key
LEFT JOIN curated.fact_medical_records m ON p.patient_key = m.patient_key
LEFT JOIN curated.fact_billing b ON p.patient_key = b.patient_key
WHERE p.is_current = TRUE
GROUP BY p.patient_key, p.patient_id, p.full_name, p.dob, p.gender, p.city;
```

```
-- Requirement 2: Risk Prediction Analysis
CREATE OR REPLACE VIEW datamart.risk_prediction_mart AS
SELECT
   p.patient_id,
   p.full_name,
   DATEDIFF('year', p.dob, CURRENT_DATE()) AS age,
   m.diagnosis,
   m.cholesterol_level,
   m.blood_pressure_systolic,
   CASE
       WHEN DATEDIFF('year', p.dob, CURRENT_DATE()) > 60 AND m.cholesterol_level > 240
AND m.blood_pressure_systolic > 140 THEN 'HIGH RISK'
            WHEN m.cholesterol_level BETWEEN 200 AND 240 OR m.blood_pressure_systolic
BETWEEN 120 AND 140 THEN 'MEDIUM RISK'
      ELSE 'LOW RISK'
   END AS health_risk_score,
   m.record_date
FROM curated.dim_patient p
JOIN curated.fact_medical_records m ON p.patient_key = m.patient_key
WHERE p.is_current = TRUE;


-- Requirement 3: Revenue Trends
CREATE OR REPLACE VIEW datamart.revenue_trends_mart AS
SELECT
   DATE_TRUNC('month', b.billing_date) AS revenue_month,
   a.department,
   COUNT(DISTINCT b.bill_id) AS total_bills_issued,
   SUM(b.total_amount) AS total_revenue,
   SUM(b.insurance_covered) AS total_insurance_claims
FROM curated.fact_billing b
JOIN curated.fact_appointments a ON b.appointment_key = a.appointment_key
GROUP BY revenue_month, a.department;


-- Requirement 4: No-Show Analysis
CREATE OR REPLACE VIEW datamart.no_show_mart AS
SELECT
   department,
   COUNT(*) AS total_appointments,
   SUM(IFF(status = 'No-Show', 1, 0)) AS total_missed_appointments,
       ROUND((SUM(IFF(status = 'No-Show', 1, 0)) * 100.0) / NULLIF(COUNT(*), 0), 2) AS
no_show_rate_percent
FROM curated.fact_appointments
GROUP BY department;
```

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-- 7. ANALYTICAL LAYER (Actionable KPIs for Dashboards)
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-- KPI 1: Enterprise Health Risk Ratio
CREATE OR REPLACE VIEW analytical.kpi_health_risk AS
SELECT
    COUNT(IFF(health_risk_score = 'HIGH RISK', 1, NULL)) AS total_high_risk_patients,
    COUNT(*) AS total_patients_analyzed,
            ROUND((COUNT(IFF(health_risk_score = 'HIGH RISK', 1, NULL)) * 100.0) / NULLIF(COUNT(*), 0), 2) AS high_risk_percentage
FROM datamart.risk_prediction_mart;

-- KPI 2: Top Performing Departments (Financial)
CREATE OR REPLACE VIEW analytical.kpi_top_departments AS
SELECT
    department,
    SUM(total_revenue) AS ytd_revenue,
    SUM(total_insurance_claims) AS ytd_insurance_claims
FROM datamart.revenue_trends_mart
GROUP BY department
ORDER BY ytd_revenue DESC;

-- KPI 3: Operational Inefficiency (No Shows)
CREATE OR REPLACE VIEW analytical.kpi_operational_inefficiency AS
SELECT
    department,
    total_missed_appointments,
    no_show_rate_percent
FROM datamart.no_show_mart
ORDER BY no_show_rate_percent DESC;

-- KPI 4: Governance Quality Score
CREATE OR REPLACE VIEW analytical.kpi_data_quality AS
SELECT
    source_table,
    error_type,
    COUNT(error_id) AS total_anomalies_quarantined
FROM governance.dq_exception_log
GROUP BY source_table, error_type
ORDER BY total_anomalies_quarantined DESC;

-- END OF SCRIPT
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-- 8. SYSTEM ALERTS (Proactive Monitoring)
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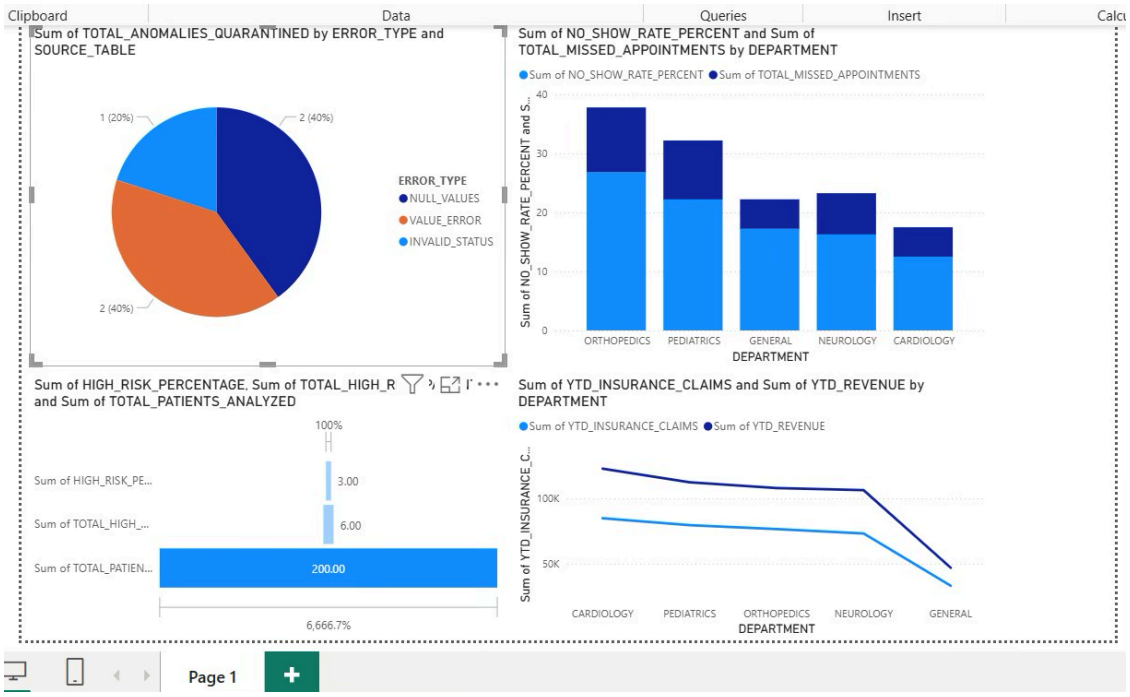
-- Create an alert to notify admins if any business anomalies are detected
CREATE OR REPLACE ALERT governance.high_anomaly_alert
  WAREHOUSE = HEALTHCARE_WH
  SCHEDULE = '10 MINUTE'
  IF (EXISTS (
      SELECT *
      FROM governance.anomaly_rule_hits
      WHERE detected_at > SNOWFLAKE.ALERT.LAST_SUCCESSFUL_SCHEDULED_TIME()
  ))
  THEN
    -- Action: Send an email (Note: Requires a pre-configured email integration 'my_email_int')
    CALL SYSTEM$SEND_EMAIL(
        'my_email_int',
        'admin@hospital.com',
        'High Anomalies Detected',
        'Check governance.anomaly_rule_hits immediately for business rule violations.'
    );

ALTER ALERT governance.high_anomaly_alert RESUME;


select * from analytical.kpi_top_departments;
select * from analytical.kpi_health_risk;
select * from analytical.kpi_operational_inefficiency;
select * from analytical.kpi_data_quality;
```

Results:


Clipboard
Data
Queries
Insert
Calcu
Sum of TOTAL_ANOMALIES_QUARANTINED by ERROR_TYPE and SOURCE_TABLE
1 (20%)
2 (40%)
2 (40%)
ERROR_TYPE
NULL_VALUES
VALUE_ERROR
INVALID_STATUS
Sum of NO_SHOW_RATE_PERCENT and Sum of TOTAL_MISSED_APPOINTMENTS by DEPARTMENT
Sum of NO_SHOW_RATE_PERCENT
Sum of TOTAL_MISSED_APPOINTMENTS
Sum of NO_SHOW_RATE_PERCENT and S...
40
30
20
10
0
ORTHOPEDICS
PEDIATRICS
GENERAL
NEUROLOGY
CARDIOLOGY
DEPARTMENT
Sum of HIGH_RISK_PERCENTAGE, Sum of TOTAL_HIGH_R and Sum of TOTAL_PATIENTS_ANALYZED
100%
Sum of HIGH_RISK_PE...
3.00
Sum of TOTAL_HIGH_...
6.00
Sum of TOTAL_PATIEN...
200.00
6,666.7%
Sum of YTD_INSURANCE_CLAIMS and Sum of YTD_REVENUE by DEPARTMENT
Sum of YTD_INSURANCE_CLAIMS
Sum of YTD_REVENUE
Sum of YTD_INSURANCE_C...
100K
50K
CARDIOLOGY
PEDIATRICS
ORTHOPEDICS
NEUROLOGY
GENERAL
DEPARTMENT
Page 1